EXPERIMENT 1

Linux File Commands and Basic Network Utilities

Aim: To practise essential Linux commands for creating directories and files, viewing and searching file contents, counting text, and checking basic network information using command-line utilities.

Requirements: A Linux terminal and, for the network checks, an active network connection. The final section also shows equivalent checks in the Windows Command Prompt.

1. Working with Directories and Files

The following commands create a working folder, confirm the current location, move into the new folder, and create a text file.

```
mkdir ThirdYearIT
pwd
cd ThirdYearIT
pwd
touch rec.txt
```

Explanation: `mkdir` creates a directory named `ThirdYearIT`. `pwd` displays the current working directory. The `cd` command enters the directory, and `touch` creates an empty file named `rec.txt`.

2. Writing to and Displaying a File

```
echo "Welcome to Third Year IT\nLinux commands are very useful." > rec.txt
echo "Welcome to Third Year IT\nLinux commands are very useful."
cat rec.txt
```

Explanation: The first command writes two lines of text into `rec.txt`; the > symbol replaces any existing content. The second command prints the text directly in the terminal. The `cat` command displays the contents saved in the file.

Expected file content:

```
Welcome to Third Year IT
Linux commands are very useful.
```

3. Searching and Counting Text

```
grep Linux rec.txt
wc -l rec.txt
wc -w rec.txt
wc -c rec.txt
wc -m rec.txt
wc -L rec.txt
```

Command	Purpose	Observed result in the supplied example
`grep Linux rec.txt`	Finds lines containing the word “Linux”.	Displays: “Linux commands are very useful.”
`wc -l rec.txt`	Counts lines in the file.	2 lines
`wc -w rec.txt`	Counts words in the file.	8 words
`wc -c rec.txt`	Counts bytes in the file.	52 bytes
`wc -m rec.txt`	Counts characters in the file.	52 characters
`wc -L rec.txt`	Shows the length of the longest line.	31 characters

Note: The values above reflect the terminal output shown in the provided document.

4. Checking System and Network Information in Linux

These commands provide information about the computer, user groups, DNS name resolution, nearby network devices, and the route taken by network traffic.

```
hostname
groups
nslookup google.com
arp -n
traceroute google.com
```

Command descriptions:

• **hostname:** Prints the system's host name. In the example, the output is `linux`.

• **groups:** Lists the groups associated with the current user.

• **nslookup google.com:** Queries DNS to find the IP address associated with the domain name. The example uses DNS server `8.8.8.8` and returns an address for Google.

• **arp -n:** Displays the ARP table, showing known local network IP addresses and their corresponding hardware (MAC) addresses.

• **traceroute google.com:** Shows the network hops between the local system and the destination. The supplied example displays four hops before reaching the destination.

5. Basic Network Checks in Windows

The third page of the source document shows similar network checks performed from Windows Command Prompt.

```
ping google.com
nslookup google.com
ping 172.16.8.1
```

Explanation: The `ping` command tests whether a destination responds over the network and reports response times. The `nslookup` command checks DNS resolution for a domain name. The final ping command tests connectivity to the local or configured network address `172.16.8.1`. In the screenshot, both ping tests receive four replies with no packet loss.

6. Result

The experiment demonstrates how to create and navigate directories, make and read text files, search file contents, count lines and words, and use common network utilities. The commands also provide basic details about the host system, DNS resolution, local network address mappings, and the route to a remote destination.

7. Conclusion

The practical exercise helped build familiarity with Linux terminal operations and introductory network troubleshooting. These commands are useful for everyday file management and for carrying out initial checks when diagnosing connectivity or name-resolution problems.